

$$TMOM_t(N) = \left(\frac{x_t}{x_{t-N}} - 1 \right) - rf$$

where

x_t	=	price of month t
N	=	number of periods
rf	=	risk-free return or the return of the Treasury bill

The example in [Table 7.2](#) outlines how the time series momentum trading rule (TMOM) would work in practice for a 12-month version of the rule ($N = 12$). Consider a hypothetical market index price of 25 and a Treasury-bill return of one percent over the past 12 months.

[Table 7.2](#) Example Time Series Momentum Calculation

Date	Price	Return
0	25	
1	24.263	−2.95%
2	24.141	−0.50%
3	24.214	0.30%
4	24.502	1.19%
5	24.384	−0.48%
6	24.716	1.36%
7	24.980	1.07%
8	24.850	−0.52%
9	24.905	0.22%
10	24.599	−1.23%
11	23.942	−2.67%
12	23.504	−1.83%
12-Month Cumulative Return		−5.99%

In the example, the past 12-month return is −5.99 percent and the Treasury bill return was 1 percent. Because the asset's return (−5.99%) is less than the